# Probabilistic Machine Learning

## Lecture 01: Introduction

Prof. Dr. Michael Granitzer

Chair of Data Science · University of Passau



# Why Probability?

**Guiding Question:** Two models both say "fraud". What would you need to know before acting on either — and why does a label not tell you?

# Machine Learning Gives Us Predictions

Three are flagged as fraud

and the three labels are identical

| | | | |
|---|---|---|---|
| Café Central<br>€4.20<br>legitimate | Supermarket<br>€63.10<br>legitimate | **Electronics Online**<br>€1,299.00<br>**fraud** | Petrol Station<br>€78.40<br>legitimate |
| **Gift Cards ×12**<br>€600.00<br>**fraud** | Bookshop<br>€24.90<br>legitimate | **Hotel Booking**<br>€410.00<br>**fraud** | Pharmacy<br>€12.75<br>legitimate |

# Machine Learning Gives Us Predictions

Three are flagged as fraud

and the three labels are identical

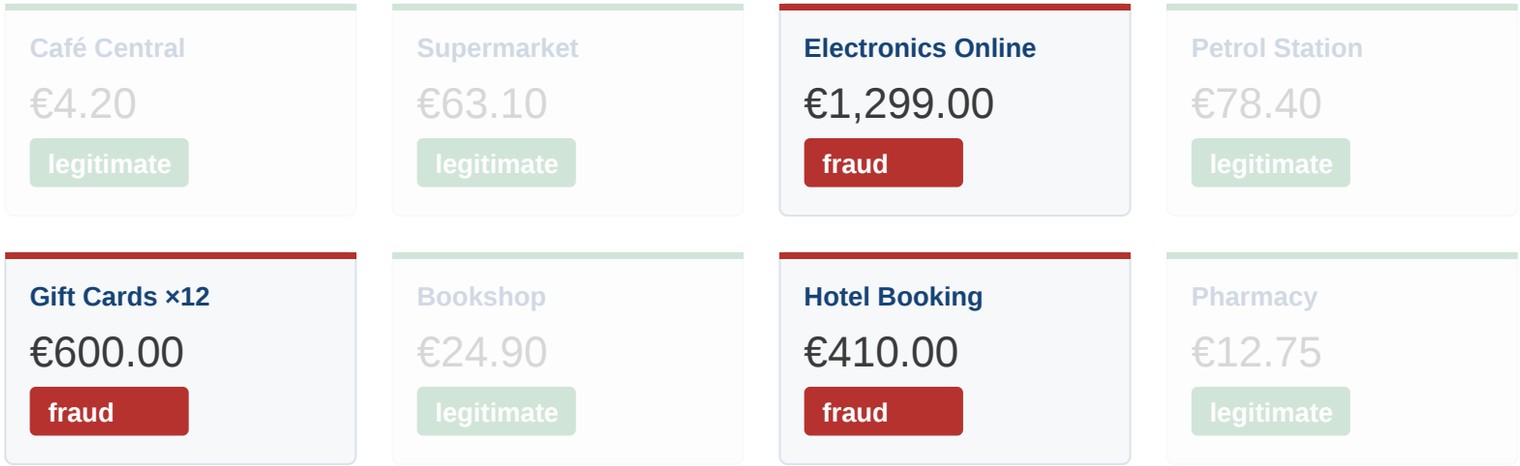
| | | | |
|---|---|---|---|
| Café Central<br>€4.20<br>legitimate | Supermarket<br>€63.10<br>legitimate | **Electronics Online**<br>€1,299.00<br>**fraud** | Petrol Station<br>€78.40<br>legitimate |
| **Gift Cards ×12**<br>€600.00<br>**fraud** | Bookshop<br>€24.90<br>legitimate | **Hotel Booking**<br>€410.00<br>**fraud** | Pharmacy<br>€12.75<br>legitimate |

Are the three flagged transactions equally suspicious?

# Machine Learning Gives Us Predictions

Three are flagged as fraud

and the three labels are identical




Are the three flagged transactions equally suspicious?

Would you block all three cards?

# Machine Learning Gives Us Predictions

Three are flagged as fraud

and the three labels are identical

| | | | |
|---|---|---|---|
| Café Central<br>€4.20<br>legitimate | Supermarket<br>€63.10<br>legitimate | **Electronics Online**<br>€1,299.00<br>**fraud** | Petrol Station<br>€78.40<br>legitimate |
| **Gift Cards ×12**<br>€600.00<br>**fraud** | Bookshop<br>€24.90<br>legitimate | **Hotel Booking**<br>€410.00<br>**fraud** | Pharmacy<br>€12.75<br>legitimate |

Are the three flagged transactions equally suspicious?

Would you block all three cards?

What would you need to know to treat them differently?



# The Same Batch, With a Number Attached




What can you do now that you could not do a minute ago?

# What a Number Lets Us Do

| The number lets us | What that means for this batch |
|---|---|
| **Compare and rank** | separate the €1,299 order from the two borderline cases — and compare two models that would both just say "fraud" |
| **Weigh consequences** | set the threshold from the cost of a missed fraud against the cost of an annoyed customer |
| **Defer deliberately** | send 0.55 and 0.51 to a human; ask whether more data would move them |
| **Combine and communicate** | merge evidence from several signals; report *how sure* we are, not only what we think |

# What a Number Lets Us Do

| The number lets us | What that means for this batch |
|---|---|
| **Compare and rank** | separate the €1,299 order from the two borderline cases — and compare two models that would both just say "fraud" |
| **Weigh consequences** | set the threshold from the cost of a missed fraud against the cost of an annoyed customer |
| **Defer deliberately** | send 0.55 and 0.51 to a human; ask whether more data would move them |
| **Combine and communicate** | merge evidence from several signals; report *how sure* we are, not only what we think |

> **We need to model uncertainty** — and we need to be able to **reason with it**.

# From Prediction to Managed Uncertainty

**Predict**
a label or a value

→

**Quantify**
how sure we are

→

**Model**
represent it explicitly

→

**Reason**
combine, compare, update

→

**Manage**
act, defer, or gather data

And once we want to manage it — **where does the uncertainty actually come from?**

> Probabilistic machine learning is not a different way of producing predictions. It is a **language for reasoning about predictions that are not certain**.

# Mini-Exercise

A loan applicant has a **default probability of 0.2**. A default costs the bank **€9,000**; rejecting a customer who would have paid costs **€1,000** in lost margin.

Accept or reject?

# Mini-Exercise

A loan applicant has a **default probability of 0.2**. A default costs the bank **€9,000**; rejecting a customer who would have paid costs **€1,000** in lost margin.

Accept or reject?

Accepting costs $0.2 \cdot 9{,}000 = €1{,}800$ in expectation; rejecting costs $0.8 \cdot 1{,}000 = €800$. **Reject —** although $p$ is far below 0.5.

# What the Threshold Tells Us

THE INTUITIVE RULE

**reject if $p > 0.5$**

accepts this applicant — whose default risk is **twice** the bank's break-even

THE RULE THE COSTS IMPLY

**reject if $p > 0.1$**

because a default costs **9×** what a lost customer does: $p^* = \frac{1{,}000}{1{,}000+9{,}000}$

The threshold comes from the **costs**, not from the probability. 0.5 is only the right cut-off when both mistakes are equally expensive — which they almost never are.

# Takeaway

A prediction without a measure of its uncertainty cannot be compared, weighed against consequences, or handed to a human at the right moment. **Modelling uncertainty and reasoning with it** is what turns a prediction into a decision.

All of this assumes the numbers are **calibrated**: a score of 0.55 is right about 55% of the time. Whether a model's output deserves that reading is a question Lecture 04 answers.

# Sources of Uncertainty

**Guiding Question:** The model says 0.55. Is that because the world is noisy, or because the model has not learned enough — and would more data help?

# Two Reasons to Be Unsure

The model says 0.55. Before asking how to *represent* that, ask why it is not 0.99.

# Two Reasons to Be Unsure

The model says 0.55. Before asking how to *represent* that, ask why it is not 0.99.

**Is the transaction itself ambiguous** — would a perfect model, with all the data in the world, still be unsure?

# Two Reasons to Be Unsure

The model says 0.55. Before asking how to *represent* that, ask why it is not 0.99.

**Is the transaction itself ambiguous** — would a perfect model, with all the data in the world, still be unsure?

**Or has the model simply not seen enough cases like this one?**

# Two Reasons to Be Unsure

The model says 0.55. Before asking how to *represent* that, ask why it is not 0.99.

**Is the transaction itself ambiguous** — would a perfect model, with all the data in the world, still be unsure?

**Or has the model simply not seen enough cases like this one?**

**Would collecting more data help — and could *any* amount of data remove the doubt entirely?**

# Two Examples

**EXAMPLE A · AN AMBIGUOUS OBSERVATION**

Two patients with **identical** records — same age, same symptoms, same test results. One recovers, one does not.

✗ Would more patients like these help? Never.

**EXAMPLE B · AN UNFAMILIAR CASE**

A house-price model trained on flats of 40–120 m² is asked about a **600 m² villa**. It has no examples anywhere near it.

✓ Would ten villas in the training data help? Completely.

Which uncertainty can be reduced, and which is part of the problem itself?

# Aleatoric vs. Epistemic Uncertainty

| | Aleatoric | Epistemic |
|---|---|---|
| **Name** | *alea*, dice — noise in the world | *episteme*, knowledge — ignorance in the model |
| **Where it comes from** | the outcome genuinely varies given what we observe | we have not seen enough to pin the model down |
| **More data?** | ✗ never — irreducible | ✓ yes — reducible |
| **Our example** | identical patients, different outcomes (A) | the 600 m² villa (B) |

# Why the Distinction Matters

| If the uncertainty is… | The right response | The expensive mistake |
|---|---|---|
| **aleatoric** (in the problem) | model the noise; set a policy for the ambiguous cases | buying more data that cannot help |
| **epistemic** (in our knowledge) | collect data where the model is thin; defer to a human who knows more | accepting it as inherent and never improving |

A model that can tell the two apart can report that it is **out of its depth**, not merely unsure.

> A probabilistic model can represent **both** kinds of uncertainty. A Bayesian one can even report them separately — noise and ignorance as two terms of one prediction (Lecture 07). That is why the distinction is worth a name.

# Takeaway

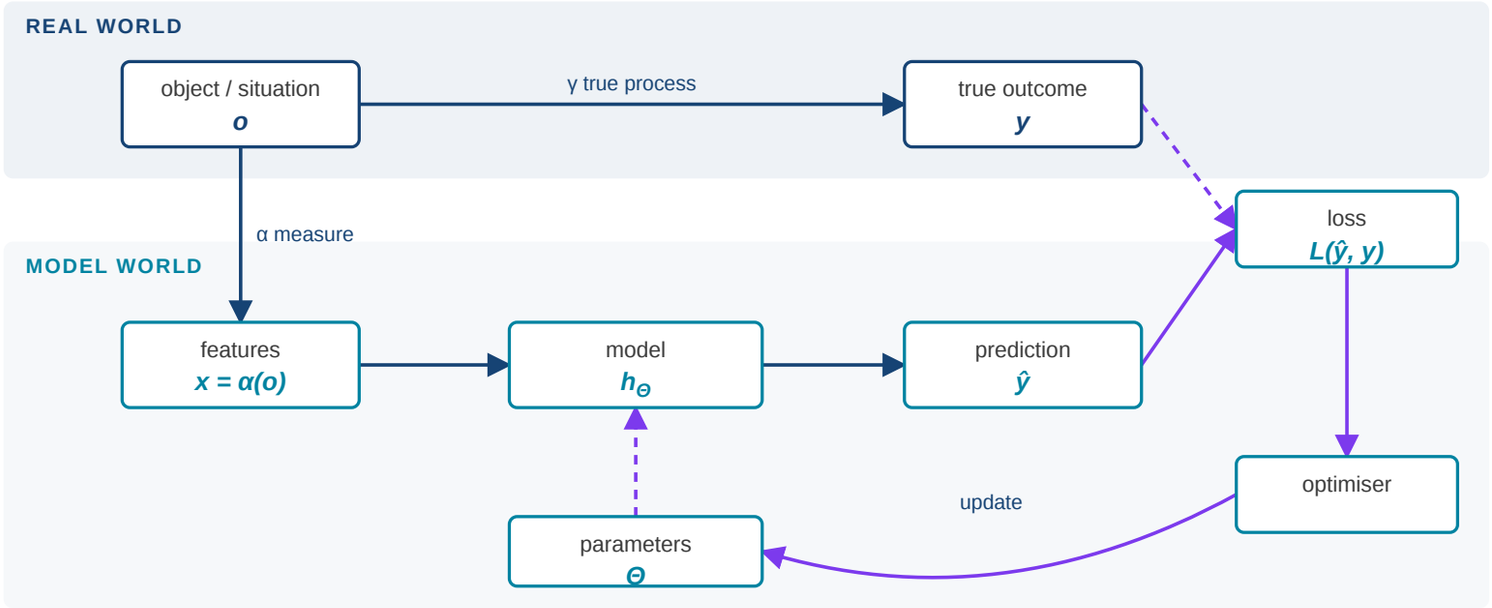
> Uncertainty has two sources: **noise in the world** and **gaps in our knowledge**. Only the second shrinks with data, and a decision — collect more, defer, act — depends on knowing which one you face.

# Uncertainty in the Learning System

**Guiding Question:** You have built learning systems before. Which of their parts were uncertain — and which of those did your model ever report?

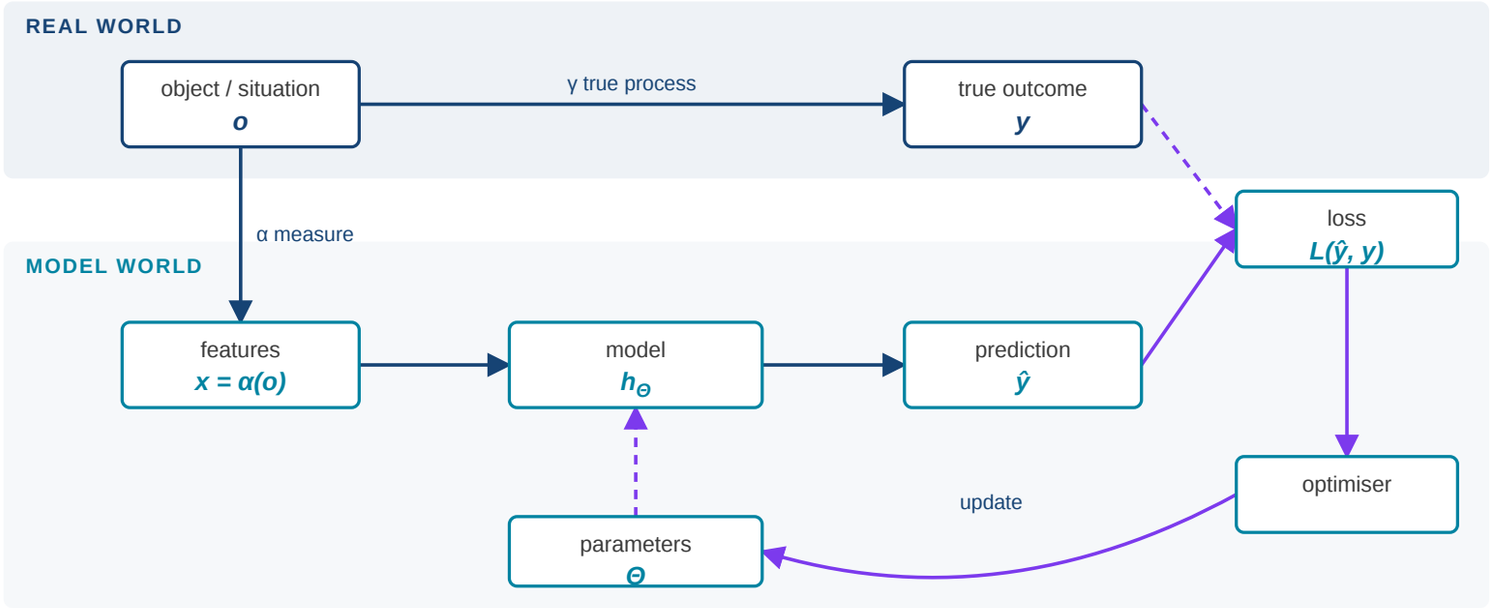
# The Learning System You Already Know

The optimiser changes the parameters — and the training loop closes



# The Learning System You Already Know

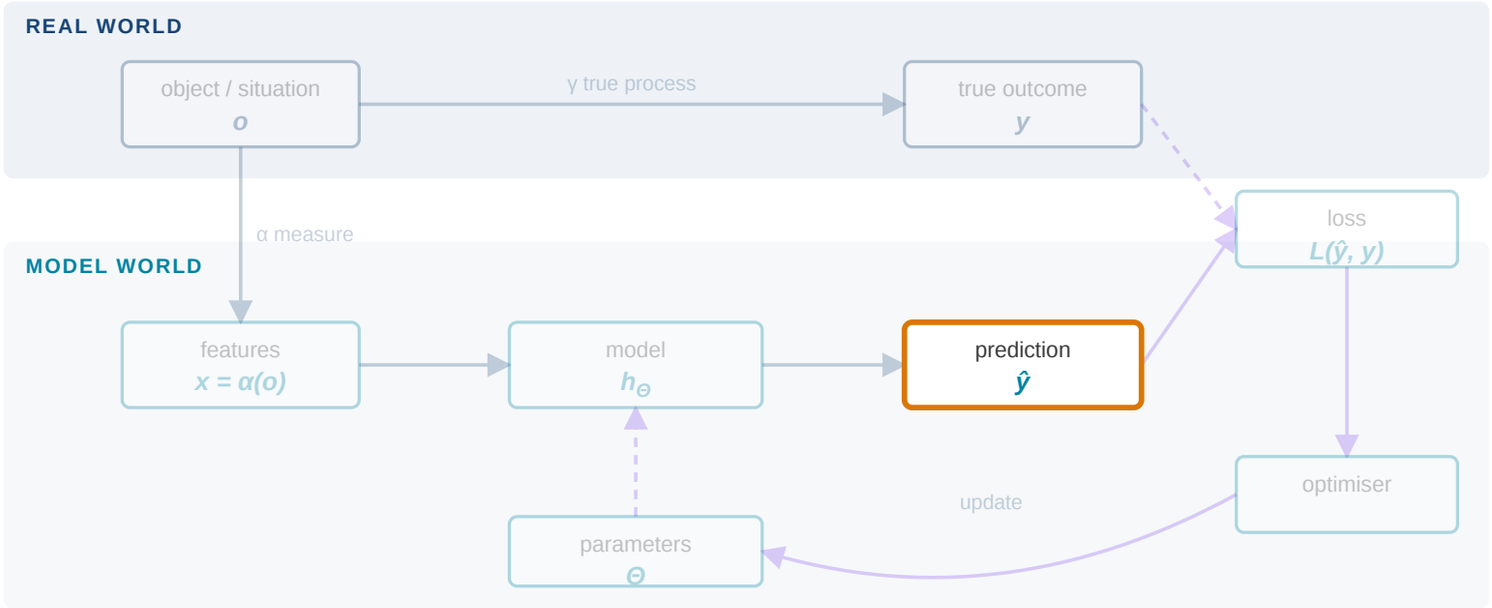
The optimiser changes the parameters — and the training loop closes




Which of these boxes did any model you have trained ever report its uncertainty about?

# The Same System, Read for Uncertainty

In the evaluation




a finite test set, and a future that may not resemble the past

# Uncertainty in vs. about the Loss

| | What is uncertain | Kind | More data? |
|---|---|---|---|
| **In the loss** | the targets $y$ are noisy | aleatoric | ✗ never |
| | the training sample is finite | epistemic | ✓ yes |
| **About the loss** | whether this objective represents the task at all | epistemic, but a modelling judgement | ✗ only knowledge of the task |

> Do not say "the loss is random". The loss is an ordinary function — say what is uncertain: the data it is computed from, or whether it is the right objective.

# Uncertainty in vs. about the Loss

| | What is uncertain | Kind | More data? |
|---|---|---|---|
| **In the loss** | the targets $y$ are noisy | aleatoric | ✗ never |
| | the training sample is finite | epistemic | ✓ yes |
| **About the loss** | whether this objective represents the task at all | epistemic, but a modelling judgement | ✗ only knowledge of the task |

Do not say "the loss is random". The loss is an ordinary function — say what is uncertain: the data it is computed from, or whether it is the right objective.

Uncertainty is not a property of the final prediction. It **enters throughout the system** — and probabilistic machine learning is the language for representing and reasoning about it at every one of these places.

# Reasoning Under Uncertainty

**Guiding Question:** A proof either holds or it does not. A learned prediction is never a proof — so what kind of conclusion is it, and how strong is it?

# Three Modes of Reasoning

| | From | To | What it guarantees |
|---|---|---|---|
| **Deduction** | premises | what **must** follow | certain — but adds no new content |
| **Induction** | observed patterns | a **regularity**, a hypothesis | nothing — content-adding, not guaranteed |
| **Abduction** | an observation | the **best available explanation** | nothing — only comparable against rivals |

Machine learning lives in the bottom two rows.

# The Cycle, on a Real Case: Uranus

**DEDUCTION**

Newton's laws plus the known planets give the orbit Uranus **must** follow.

**INDUCTION**

Six decades of observed positions show a **systematic** deviation — larger than any measurement error.

**ABDUCTION**

Best explanation: an **unseen planet** further out, pulling on Uranus.

# The Cycle, on a Real Case: Uranus

**DEDUCTION**

Newton's laws plus the known planets give the orbit Uranus **must** follow.

**INDUCTION**

Six decades of observed positions show a **systematic** deviation — larger than any measurement error.

**ABDUCTION**

Best explanation: an **unseen planet** further out, pulling on Uranus.

The cycle turns: the abduced planet becomes a new premise. Le Verrier deduces where it must be; in 1846 Galle points a telescope there and finds **Neptune**, about a degree from the prediction.

# An Anomaly Leaves Two Moves

SAVE THE THEORY

## Add a hidden cause

*The bet:* the theory is right; our picture of the world is incomplete.

*If wrong:* we chase something that is not there.

CHANGE THE THEORY

## Replace the premises

*The bet:* the theory fails here, however well it worked elsewhere.

*If wrong:* we throw away two centuries of correct predictions.

1859: Mercury's orbit turns faster than Newton allows — about **43″ per century** by the modern figure. Which move would you make?

# Same Move, Opposite Outcomes

| anomaly | the move | what happened |
|---|---|---|
| **Uranus**, 1840s | a hidden planet | **Neptune** found, 1846 ✓ |
| **Mercury**, 1859 | a hidden planet, "Vulcan" (Le Verrier again) | ? |

# Same Move, Opposite Outcomes

| anomaly | the move | what happened |
|---|---|---|
| **Uranus**, 1840s | a hidden planet | **Neptune** found, 1846 ✓ |
| **Mercury**, 1859 | a hidden planet, "Vulcan" (Le Verrier again) | ? |

Vulcan is **never found** ✗ In 1915 **general relativity** explains the 43″ with no new planet — here, changing the theory was right.

# Same Move, Opposite Outcomes

| anomaly | the move | what happened |
|---|---|---|
| **Uranus**, 1840s | a hidden planet | **Neptune** found, 1846 ✓ |
| **Mercury**, 1859 | a hidden planet, "Vulcan" (Le Verrier again) | ? |

Vulcan is **never found** ✗ In 1915 **general relativity** explains the 43″ with no new planet — here, changing the theory was right.

> Both decisions were **reasonable when they were made**. What separated them was evidence that did not exist yet.

# Deciding Before the Evidence Is In

- → **Strength of belief.** After two centuries of success, absorbing one anomaly *into* the theory was the rational first move.
- → **Evidence accumulates.** Every failed search made "a hidden planet" less plausible — belief moved step by step, not in one jump.
- → **Replace, don't just refute.** Newton fell when a rival explained *more*, not at the first failure.
- → **Costs decide.** Chasing Vulcan forever and abandoning a good theory too early are both expensive; which move to make depends on what being wrong costs.

> Beliefs of different strength, updated by evidence, acted on under costs — the whole lecture in one historical case. Lecture 03 gives each of these words its mathematics.

# Mini-Exercise — Your Model Fails on a Subgroup

A fraud model misses most fraud from a **newly added country**. You add 500 more labelled transactions from there, retrain — and the error **does not move**.

Noise, too little data, or the wrong model?

# Mini-Exercise — Your Model Fails on a Subgroup

A fraud model misses most fraud from a **newly added country**. You add 500 more labelled transactions from there, retrain — and the error **does not move**.

Noise, too little data, or the wrong model?

**Not too little data —** then the error would have dropped. Left: the wrong model (try new features or a richer model) or noise (nothing helps, and experts with the same information cannot tell either). You saved the theory once; now the evidence says to change it.

# Takeaway

> Learning from data never proves; it **bets**. An anomaly can be absorbed or can overturn the model, and which move is right is decided **under uncertainty** — by how strong the belief is, how the evidence accumulates, and what being wrong costs.

# Beyond True or False

| | Classical reasoning | Probabilistic reasoning |
|---|---|---|
| **Conclusions** | follow, or do not | are supported to *degrees* |
| **Evidence** | premises, taken as given | may be partial or noisy |
| **Certainty** | the conclusion is certain | the conclusion may stay uncertain |
| **New information** | can only add premises | can *change* what we believe |

Classical logic asks *whether* a conclusion follows. Probabilistic reasoning also asks **how strongly the evidence supports each possible conclusion**.

# What the Language Has to Do

We now know that uncertainty matters and that it has different sources. What we still lack is a **systematic way** to:

1. **Represent** — say how much we believe each possibility
2. **Combine** — put several uncertain pieces of information together
3. **Update** — change our beliefs when new evidence arrives
4. **Decide** — act on beliefs that are not certainties

# What the Language Has to Do

We now know that uncertainty matters and that it has different sources. What we still lack is a **systematic way** to:

1. **Represent** — say how much we believe each possibility
2. **Combine** — put several uncertain pieces of information together
3. **Update** — change our beliefs when new evidence arrives
4. **Decide** — act on beliefs that are not certainties

> **Probability provides that language.** For representing, combining and updating there is essentially one consistent set of rules; deciding adds a second ingredient — what the outcomes cost.

# Takeaway

Machine-learning predictions are inferences from evidence, not derivations from premises. **Probability is the calculus of such inferences** — which is why Lecture 02 is about it, and why it will feel like an answer rather than a prerequisite.

# What Probabilistic ML Enables

**Guiding Question:** Which questions about a machine-learning system will you be able to answer after this course that you cannot answer today?

# After This Course, You Will Be Able to Ask — and Answer

### ABOUT A PREDICTION

- ? How certain is it, and should I trust it?
- ? When should the model refuse to decide?
- ? What alternatives are plausible?

### ABOUT THE DATA

- ? When would more data help?
- ? When is uncertainty unavoidable?
- ? What does this data actually justify?

### ABOUT LEARNING

- ? How should belief change when evidence arrives?
- ? How do I combine prior knowledge with observations?
- ? How do I compare competing explanations?

### ABOUT THE METHODS

- ? Why do the common methods work, and under which assumptions?
- ? When should I expect them to fail?
- ? What is my model assuming without telling me?

# What Changes Compared to Traditional Machine Learning

**TRADITIONAL ML ASKS**

**What is the prediction?**

**PROBABILISTIC ML ALSO ASKS**

- ? **How certain** is it?
- ? **What alternatives** are plausible?
- ? **Why** are we uncertain?
- ? **How does new evidence** change what we believe?
- ? **What decision** follows from this?
- ? **What can and cannot** be inferred from this information?

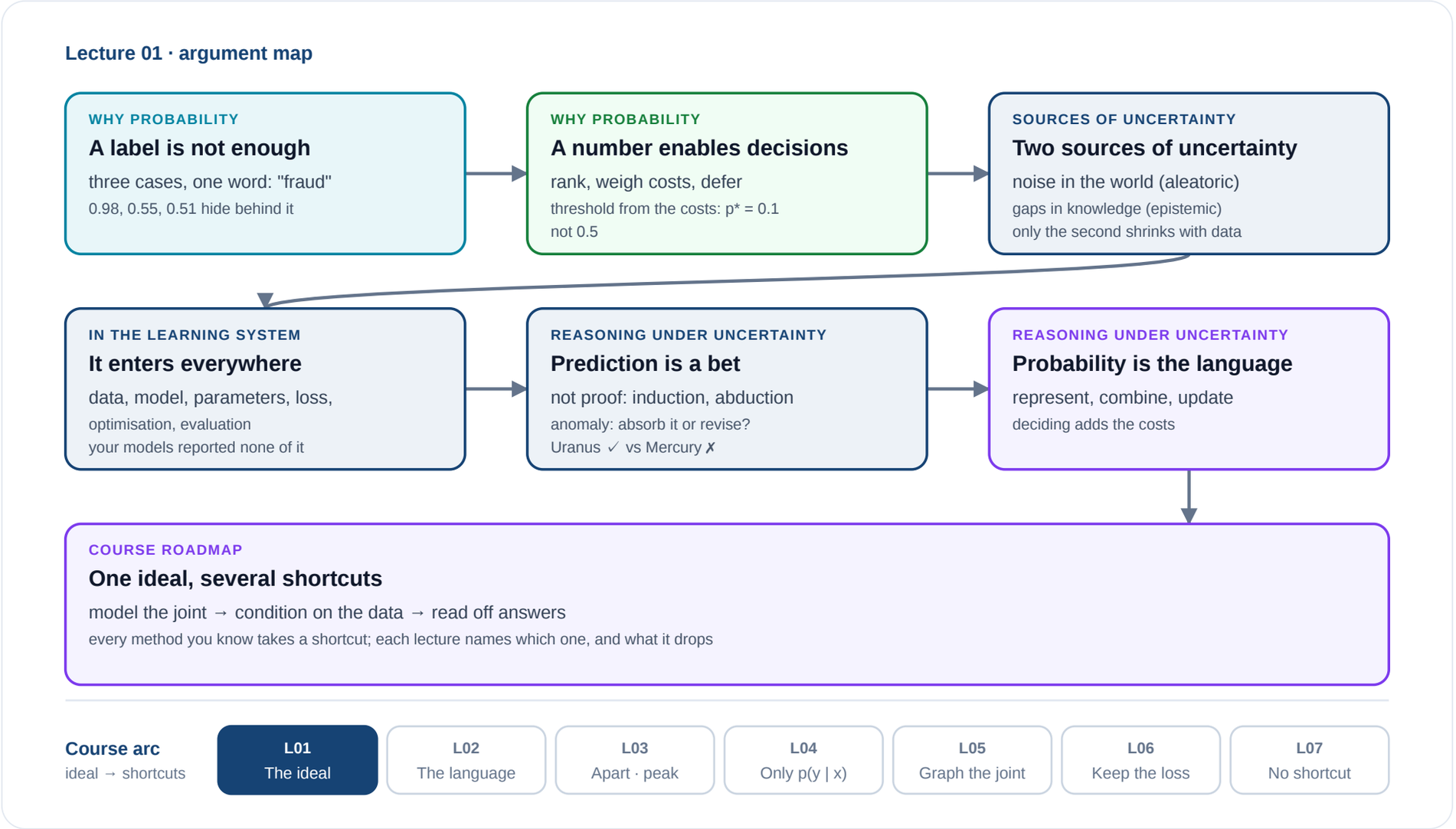
# Takeaway

> Probabilistic machine learning does not replace prediction. It **surrounds every prediction with the questions a decision needs answered** — and gives one language in which all of them can be asked.

# Course Roadmap

**Guiding Question:** Now that the motivation is in place — which lecture answers which of your questions?

# Argument Map



# The Road Ahead

Each lecture answers one of the questions you have just met.

## 02

**Representing uncertainty** *How do we describe uncertainty mathematically?*

Probability, distributions, and how to summarise data.

## 03

**Generative models and naive Bayes** *What does a classifier look like when it is built straight from Bayes — and what does it cost?*

Keeping the peak, modelling the joint $p(x, y)$, naive Bayes.

## 04

**Discriminative models** *What do linear and logistic regression assume — and when is their output a probability?*

Modelling only $p(y \mid x)$: linear and logistic regression, calibration.

## 05

**Structured models** *How do we write down a large, structured set of beliefs?*

Graphical models: drawing the assumptions, fitting them, computing with them.

## 06

**Judging models** *What is the quantity we actually want to minimise — and what did we get instead?*

Evaluation and empirical risk minimisation.

## 07

**Deciding and refining** *How do we keep uncertainty all the way to the decision?*

Mixtures, Bayesian regression, and what comes after.

# The Ideal, in Three Moves

Every method in this course is measured against the same three moves, in the language you are about to learn:

### Model the joint

Write down what the world would have to be like to produce what we observed — the data *and* the unknowns together.

### Condition on the data

Inference is conditioning: fix what was observed, and let the rules of probability say what follows for the rest.

### Read off answers

Predictions, uncertainties and decisions are all quantities computed from that one conditioned model.

> Every one of these words — joint, condition, inference — gets a precise meaning in Lecture 02. They appear here only so that you recognise them when they return.

# …and the Shortcuts Everyone Takes

Most methods you already know take a shortcut — and each one drops something.

| Shortcut | Instead | What is lost | L |
|---|---|---|---|
| **Keep the peak** | a single best parameter (maximum likelihood or MAP) | how sure we are about it | 03–04 |
| **Assume it apart** | factorise the joint by independence | the dependences assumed away | 03 |
| **Only what you need** | model $p(y \mid x)$, never the joint (discriminative) | every other question | 04 |
| **Graph the joint** | state exactly which independences hold | cheap inference, in general | 05 |
| **Keep the loss** | minimise average error on the sample (ERM) | every probability | 06 |
| **No shortcut** | keep the whole posterior — EM only for the hidden labels | computation; it cannot see a wrong model family | 07 |

> Knowing which shortcut a method takes tells you which questions it can no longer answer.